

INF-001 - Syrka Capability Inference Constitution

Status: Constitutional Engineering Specification

Version: 1.0

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Builds upon: Product Suite; ADR-001 - Open Source LMS Foundation; ADR-002 - Domain Architecture & Bounded Contexts; CEA-001 - Canonical Event Architecture; ONT-001 - Ontology Constitution; GRAPH-001 - Human Capability Graph Specification; REASON-001 - Evidence & Reasoning Constitution
Decision owners: AI Science / Knowledge Engineering / Ontology / ML Architecture / Platform Architecture / Research

Scope: Capability definitions, lifecycle, inference pipeline, evidence aggregation, evolution, dependencies, similarity, transfer, decay, verification, AI independence, human oversight, explainability, examples, and governance

1. Executive Overview

Capability inference is Syrka's central intelligence layer. It is the process that transforms educational activity, assessments, projects, research, professional evidence, credentials, employer signals, and canonical events into evidence-backed capability assertions that can power learning recommendations, career intelligence, employer matching, workforce planning, government analytics, and the Syrka Job Passport.

Capability inference is distinct from adjacent layers:

| Layer | Responsibility | What it does not do |

|---|---|---|

| Ontology | Defines what concepts mean. | Does not decide whether a person has a capability. |

| Evidence & Reasoning | Defines how Syrka reasons from evidence. | Does not own capability lifecycle state. |

| Human Capability Graph | Represents semantic relationships over time. | Does not independently invent conclusions. |

| Capability Inference | Converts evidence into capability observations, confidence updates, recommendations, and passport-eligible claims. | Does not redefine ontology or bypass reasoning rules. |

| AI models | Assist extraction, classification, mapping, scoring, and explanation. | Are not constitutional sources of truth. |

The inference layer transforms educational data into workforce intelligence by converting raw learning and activity events into portable, context-aware, evidence-backed capability claims. A course completion is not a capability. A grade is not a capability. A GitHub repository is not a capability. A capability emerges only when evidence is mapped to ontology concepts, evaluated for relevance and quality, placed in graph context, scored for confidence, checked for contradiction, and published as a canonical capability event.

Constitutional premise:

Evidence + Ontology + Graph Context + Reasoning Rules + Confidence Model + Review Policy
-> Capability Observation / Capability Update / Recommendation / Passport Claim

2. Capability Definition

| Term | Canonical definition | Example | Counterexample |

|---|---|---|---|

| Capability | Evidence-backed, context-aware ability to perform actions, solve problems, produce outcomes, or exercise judgment at stated confidence. | Can design normalized relational database schemas in academic project context. | "Completed Database 101" without assessed evidence. |

| Capability Observation | A temporal evidence-backed observation that a subject demonstrates or may demonstrate a capability. | CapabilityObserved from graded schema project. | A static skill label on a profile. |

| Capability Assertion | Accepted claim that a subject has a capability in a context with confidence/provenance. | Passport claim for SQL data modeling at Strong confidence. | LLM free-text claim with no evidence. |

| Capability State | Current graph projection of capability confidence, evidence, validity, and status. | active, strong, verified, stale, revoked. | Raw event stream. |

| Capability Confidence | Calibrated support level for a capability assertion under uncertainty. | 0.84 Strong. | Grade percentage copied as confidence. |

| Capability Strength | Practical robustness of capability across difficulty, contexts, and evidence diversity. | Strong across coursework, project, and internship. | One easy quiz result. |

| Capability Context | Conditions under which capability is believed to hold. | academic, production, clinical, regulated jurisdiction. | Universal claim without boundaries. |

| Capability Freshness | Current relevance of evidence supporting a capability. | Cloud certification renewed this year. | Five-year-old framework knowledge in obsolete version. |

| Capability Evolution | Change in capability state over time. | Emerging -> Supported -> Strong. | Overwriting old state. |

| Capability Decay | Reduction in confidence due to time, disuse, obsolescence, contradiction, or context shift. | Legacy AngularJS skill decays for modern frontend roles. | Deleting prior evidence. |

| Capability Revocation | Invalidity or removal of current positive support due to revoked evidence, fraud, policy, or review. | Revoked credential removes passport support. | Hiding inconvenient evidence. |

| Capability Transferability | Degree to which capability in one context supports another context. | Military logistics supports supply-chain operations. | Assuming all leadership transfers equally. |

| Capability Similarity | Semantic closeness between capabilities based on ontology, evidence, tasks, tools, and outcomes. | Python and Java share programming foundations but differ ecosystems. | Treating Python and accounting as similar due to same course. |

| Capability Hierarchy | Parent/child organization of capabilities. | Data engineering contains pipeline design. | Job title hierarchy. |

| Capability Dependency | Required or supporting capability relationship. | Machine learning requires statistics and programming. | Skill popularity correlation. |

| Capability Composite | Capability composed of multiple capabilities, skills, knowledge, and behaviors. | Product management combines research, prioritization, communication. | Single isolated task. |

| Capability Gap | Difference between observed capability state and required target state. | Missing clinical license for residency pathway. | Low self-esteem. |

| Capability Potential | Plausible future capability under defined support and learning conditions. | High potential for AI after math/programming evidence. | Verified capability. |

| Capability Readiness | Degree to which capability/profile satisfies requirements for action, role, credential, or mobility route. | Ready for junior data analyst role. | Interest in data science. |

| Capability Risk | Risk that a capability claim is overstated, stale, biased, context-mismatched, or unsupported. | High risk from self-report-only claim. | Low confidence itself without reason. |

| Capability Verification | Confirmation of evidence/claim by trusted process, issuer, employer, expert, or government. | Professional license verified. | Course page viewed. |

| Capability Validation | Check that inference conforms to ontology, evidence rules, schemas, thresholds, and policy. | Schema and rule validation before CapabilityObserved. | Model output accepted directly. |

3. Capability Lifecycle

Observation

v

Evidence

v

Hypothesis

v

Candidate Capability

v

Confidence Evaluation
 v
 Conflict Resolution
 v
 Capability Assertion
 v
 Capability Evolution
 v
 Capability Strengthening
 v
 Capability Decay
 v
 Capability Revocation
 v
 Historical Archive

Stage	Meaning	Events/graph objects	Rule
Observation	Syrka captures something that happened.	canonical events, observation nodes	Not a conclusion.
Evidence	Observation/artifact becomes relevant to a claim.	Evidence node, DERIVED_FROM	Must have provenance.
Hypothesis	System proposes possible capability.	candidate inference object	Not user-facing as verified claim.
Candidate Capability	Hypothesis mapped to ontology concept/context.	capability concept, context	Requires ontology version.
Confidence Evaluation	Evidence is scored and aggregated.	ConfidenceScore	Must be calibrated and explainable.
Conflict Resolution	Contradictions are represented and evaluated.	CONTRADICTS, review flags	Conflicts never hidden.
Capability Assertion	Accepted capability observation is published.	CapabilityObserved, HCG assertion	Requires evidence and provenance.
Capability Evolution	State changes with new evidence.	CapabilityUpdated, supersession edges	Append-only history.
Capability Strengthening	Confidence/strength increases through evidence.	updated confidence	Requires non-redundant support.
Capability Decay	Confidence decreases due to staleness/change.	decay event/update	Capability-specific decay rules.
Capability Revocation	Current assertion invalidated.	revocation/correction event	History preserved.
Historical Archive	Past state retained for audit/explanation.	snapshots/events	Reproducible from event store.

4. Inference Pipeline

4.1 Inputs

Input	Purpose
Canonical Events	Immutable facts such as AssignmentGraded, GitHubRepositoryLinked, CredentialVerified.
Human Capability Graph	Existing capability, evidence, ontology, occupation, and opportunity context.
Ontology	Canonical meaning of capabilities, dependencies, similarity, hierarchy, and standards.
Evidence	Validated artifacts, assessments, credentials, reviews, research, projects, work signals.
Market Signals	Demand, salary, occupation requirements, emerging skills, disruption signals.
Career Intent	User goals, preferences, geography, industry, salary, constraints.
Human Review	Faculty, employer, peer, expert, government, or student challenge outcomes.

4.2 Outputs

Output	Meaning
---	---
Capability Assertions	Accepted evidence-backed claims with context/confidence/provenance.
Confidence Updates	Changes to capability confidence, strength, decay, or risk.
Recommendations	Learning, career, evidence collection, opportunity, or mobility suggestions.
Job Passport Claims	Portable claims that meet passport inclusion thresholds and disclosure policy.
Learning Path Suggestions	Actions to close capability gaps or increase confidence.

4.3 Pipeline stages

1. Ingest canonical event
2. Validate event and evidence references
3. Extract evidence features
4. Map evidence to ontology concepts
5. Retrieve graph neighborhood
6. Generate capability hypotheses
7. Score evidence and confidence
8. Detect conflicts and decay
9. Apply policy/human-review gates
10. Publish capability events
11. Update graph projections
12. Trigger recommendations/passport updates

No stage may bypass canonical events, ontology, graph provenance, or reasoning rules.

5. Evidence Aggregation

5.1 Combination rules

Pattern	Effect on confidence
---	---
Multiple independent strong evidence items	Significant confidence increase.
Repeated identical weak signals	Small increase, quickly saturates.
Duplicate evidence from same source	Deduplicated or heavily discounted.
Evidence across contexts	Increases transferability and strength.
Evidence covering all required sub-capabilities	Increases composite capability confidence.
Missing required component	Caps confidence for composite capability.
Strong contradiction	Reduces confidence and may trigger review.
Revoked evidence	Removed from positive support; may create contradiction.
Expired evidence	Supports history, weakens current confidence.
Refreshed evidence	Restores or increases freshness/confidence.

5.2 Aggregation formula family

A capability confidence function must be versioned and reproducible:

```
C(capability, subject, context, t) = aggregate(
  weighted_supporting_evidence,
  weighted_conflicting_evidence,
  required_component_coverage,
  source_trust,
```

```
evidence_independence,
freshness,
context_match,
calibration_adjustment,
human_review_adjustment
)
```

Aggregation is not a naive average. It must account for the weakest required component, evidence diversity, contradiction, and uncertainty.

5.3 Evidence expiry and refresh

- Evidence expiry is capability-specific and jurisdiction-specific where required.
- Expiry does not delete historical capability; it lowers current confidence.
- Refresh occurs when new evidence confirms continued capability in a relevant context.
- Refreshed evidence must be independent enough to avoid artificial confidence inflation.

6. Capability Evolution

Evolution driver	Effect	Example
Practice	Increases confidence/strength if practice is relevant and non-trivial.	repeated clinical simulations.
Feedback	Improves capability when followed by better evidence.	rubric feedback then improved project.
Work experience	Strengthens context-specific capability.	production incidents resolved.
Research output	Strengthens research/analytical capability.	publication with clear contribution.
Forgetting/disuse	Causes decay when capability requires active practice.	unused language fluency.
Technological change	Obsoletes or narrows capability.	old framework version.
Industry evolution	Changes capability relevance or requirements.	AI changes data analyst role.
Credential renewal	Refreshes verified support.	safety certification renewed.
Revocation/fraud	Reduces or invalidates support.	plagiarized project discovered.

Capability maturity bands:

Maturity	Meaning
Exposed	Person has encountered concept; no performance evidence.
Emerging	Early evidence exists; limited context.
Developing	Multiple evidence items; gaps remain.
Proficient	Reliable performance in defined context.
Advanced	Strong performance across varied contexts.
Expert	Sustained high-confidence capability with transfer, judgment, and recognition.
Stale	Historical evidence exists but current confidence decayed.
Revoked	Positive assertion invalidated for current use.

7. Capability Dependencies

Dependency type	Definition	Rule
Prerequisite capability	Required before target can be credibly performed.	Caps target confidence if missing.
Supporting capability	Increases likelihood/quality but not strictly required.	Adds support but does not cap alone.
Composite capability	Capability composed of multiple capabilities/skills/knowledge.	Confidence depends on coverage and weakest critical component.
Transferable capability	Capability reusable across contexts.	Requires context similarity and transfer evidence.

|
Specialized capability	Capability tied to narrow domain/tool/jurisdiction.	Transfer is limited.
General capability	Broad capability applicable across many contexts.	Requires diverse evidence to avoid vagueness.
Hidden capability	Capability suggested indirectly by evidence.	Must remain hypothesis until direct/strong support exists.
Latent capability	Potential capability not yet demonstrated.	Cannot be passport claim; can guide recommendations.

Dependency rules:

1. Required dependencies may cap confidence.
2. Supporting dependencies may increase confidence but cannot prove target alone.
3. Composite capabilities require explicit component coverage.
4. Latent/hidden capabilities must not be represented as verified capability.
5. Dependency graphs are ontology-versioned.

8. Capability Similarity

Similarity is semantic, not label-based. It uses ontology distance, shared prerequisites, shared tools, task overlap, evidence transfer, industry context, and historical transition outcomes.

Pair	Similarity rationale	Transfer caution
Python ? Java	Shared programming concepts, OOP, debugging, software design.	Syntax/ecosystem/tooling differ.
Statistics ? Econometrics	Shared probability, regression, inference.	Econometrics has domain assumptions and causal methods.
Leadership ? Team Management	Shared coordination, communication, accountability.	Leadership scope may exceed team operations.
OSINT ? Intelligence Analysis	Shared evidence collection, source evaluation, inference.	Legal/security context differs.
Urban Planning ? Urban Regeneration	Shared spatial, policy, community, infrastructure concepts.	Regeneration includes finance, redevelopment, stakeholder delivery.

Similarity metrics:

$$\text{similarity} = \text{ontology_overlap} + \text{prerequisite_overlap} + \text{task_overlap} + \text{tool_overlap} + \text{evidence_transfer_rate} + \text{market_transition_success} - \text{context_penalty}$$

Similarity must produce explanation, not only score.

9. Capability Transfer

Capability transfer is inference that evidence in one context supports capability in another context.

Transfer example	Transferable elements	Required caution
Military logistics -> Supply Chain	planning, coordination, inventory, risk, operations.	civilian systems, commercial metrics, software tools.
Academic research -> Market Research	literature review, methods, data analysis, writing.	customer/business context and speed.
Software Engineering -> Quantitative Development	programming, algorithms, testing, systems thinking.	finance

math, risk, low-latency or quant methods. |
 | Political Science -> Geopolitical Risk | institutions, policy analysis, qualitative reasoning. | market/security forecasting and data sources. |
 | Civil Engineering -> Infrastructure Consulting | infrastructure systems, project constraints, regulation. | client advisory, finance, strategy consulting. |

Transfer rules:

1. Transfer never creates verified capability by itself.
2. Transfer creates hypotheses or lower-confidence support unless direct evidence exists.
3. Transfer confidence decays with context distance.
4. Transfer requires ontology path and evidence path.
5. High-impact transfer claims require human/expert review.

10. Capability Decay

Decay type	Trigger	Effect
Time decay	Evidence age exceeds freshness window.	Confidence decreases gradually or sharply by capability.
Practice decay	No recent performance/practice.	Practical capability weakens.
Knowledge obsolescence	Domain knowledge changes.	Older evidence loses relevance.
Certification expiry	Credential expires.	Verified support removed for current claims.
Industry disruption	Market requirements change.	Context match decreases.
Technology shift	Tools/frameworks/languages change.	Tool-specific capability decays.
Skill atrophy	Human performance declines without use.	Confidence and readiness decrease.
Context change	New jurisdiction/role/domain.	Transferability and readiness decrease.

Decay rule:

$$\text{current_confidence} = \text{prior_confidence} \times \text{freshness_factor} \times \text{practice_factor} \times \text{context_factor} \times \text{technology_relevance} \times \text{contradiction_factor}$$

Decay must be explainable and reversible through refreshed evidence.

11. Capability Verification

Verification strength ranking from weakest to strongest by default:

Rank	Verification type	Strength	Notes
1	Self-reported capability	Very weak	Can seed hypotheses only.
2	Behavioral/passive evidence	Weak	Attendance/views/discussion need caution.
3	Portfolio-backed capability	Medium	Depends on authenticity and review.
4	Assessment-backed capability	Medium-strong	Depends on assessment validity and difficulty.
5	Peer-reviewed capability	Medium-strong	Depends on peer trust and bias controls.
6	Research-backed capability	Strong	Requires authorship/contribution clarity.
7	Employer-verified capability	Strong	Context-specific; bias checks required.
8	Government-recognized capability	Very strong	for regulated context Jurisdiction-specific.
9	Professional-license capability	Very strong	when current Strong for regulated practice; may not cover all capabilities.

Verification is claim-specific. A professional license may strongly verify legal eligibility but only partially support a specific technical capability.

12. AI Independence

AI providers may participate in inference by extracting evidence, mapping ontology concepts, proposing hypotheses, summarizing artifacts, detecting conflicts, estimating similarity, or drafting explanations. Providers remain subordinate to this constitution.

Rules:

1. GLM, Claude, GPT, Gemini, and future models are interchangeable adapters.
2. Model output is candidate evidence/reasoning, not accepted capability.
3. Model-specific confidence is not Syrka confidence.
4. Every model output must cite evidence IDs and ontology concept IDs.
5. Provider replacement requires regression tests on capability inference benchmarks.
6. No downstream passport, graph, or recommendation service may require provider-specific opaque fields.
7. AI-derived high-impact claims require policy gates and possibly human review.

13. Human Oversight

Oversight type	Purpose	Constitutional rule
Faculty review	Validate educational evidence and assessment interpretation.	Review event required; cannot erase history.
Employer review	Validate workplace evidence and role performance.	Must record employer context and potential bias.
Student appeal	Allow challenge of evidence/inference/passport claim.	Must provide resolution and graph update.
Government review	Validate regulated mobility/credential recognition.	Jurisdiction overlay applies.
Expert adjudication	Resolve domain-specific ambiguity.	Expert identity and rationale recorded.
Capability correction	Correct mapping, evidence, or confidence error.	Publish correction/supersession event.
Capability revocation	Invalidate current claim.	Preserve historical assertion with revoked status.
Human override	Override model/rule output.	Must include rationale and reviewer authority.

Oversight preserves provenance by creating new canonical events and graph assertions. It never deletes inconvenient history.

14. Explainability

Every inferred capability must answer:

Question	Required content
What evidence created it?	Evidence IDs, artifacts, assessments, credentials, reviews.
What evidence contradicts it?	Contradictory evidence IDs and handling rationale.
Why does Syrka believe it exists?	Reasoning summary, evidence weights, ontology mapping.
What ontology concepts were used?	Concept IDs and ontology version.
Which graph paths were traversed?	HCG path summary or query trace.
What confidence exists?	Score, band, model/rule version, calibration status.
What uncertainty remains?	Missing evidence, stale evidence, context limitations, model disagreement.
What review history exists?	Human review, appeal, correction, revocation events.

Minimum capability inference record:

{

```

"capability_observation_id": "cap_obs_001",
"subject_id": "person_123",
"capability_id": "capability:sql_data_modeling",
"context": "academic database project",
"confidence": { "score": 0.84, "band": "Strong", "model": "confidence@1.0.0" },
"evidence_ids": ["evidence_assignment_789", "artifact_schema_project"],
"contradictions": [],
"ontology_version": "1.0.0",
"graph_paths": ["Person->CapabilityObservation->Evidence->CanonicalEvent"],
"source_event_ids": ["AssignmentSubmitted:abc", "AssignmentGraded:def"],
"uncertainty": ["academic context, not production setting"],
"review_history": []
}

```

15. Mermaid Diagrams

15.1 Capability lifecycle

stateDiagram-v2

```

[*] --> Observation
Observation --> Evidence
Evidence --> Hypothesis
Hypothesis --> CandidateCapability
CandidateCapability --> ConfidenceEvaluation
ConfidenceEvaluation --> ConflictResolution
ConflictResolution --> CapabilityAssertion
CapabilityAssertion --> CapabilityEvolution
CapabilityEvolution --> Strengthening
CapabilityEvolution --> Decay
Decay --> Revocation
Strengthening --> HistoricalArchive
Revocation --> HistoricalArchive

```

15.2 Inference pipeline

flowchart TB

```

Events[Canonical Events] --> Evidence[Evidence Extraction]
Evidence --> Validation[Evidence Validation]
Validation --> Ontology[Ontology Mapping]
Ontology --> Graph[Graph Neighborhood Retrieval]
Graph --> Hypothesis[Capability Hypotheses]
Hypothesis --> Scoring[Confidence Scoring]
Scoring --> Conflicts[Conflict Detection]
Conflicts --> Review{Human review required?}
Review -- yes --> HumanReview[Human Review]
Review -- no --> Publish[Publish CapabilityObserved]
HumanReview --> Publish
Publish --> HCG[Human Capability Graph]
Publish --> Recommendations
Publish --> Passport[Job Passport]

```

15.3 Capability evolution

flowchart LR

Emerging --> Developing
Developing --> Proficient
Proficient --> Advanced
Advanced --> Expert
Proficient --> Stale
Stale --> Refreshed
Refreshed --> Proficient
Stale --> Revoked

15.4 Evidence aggregation

flowchart TB

EvidenceA[Assessment Evidence] --> Weighting
EvidenceB[Portfolio Evidence] --> Weighting
EvidenceC[Employer Evidence] --> Weighting
EvidenceD[Research Evidence] --> Weighting
Contradiction[Conflicting Evidence] --> Weighting
Weighting --> Aggregation
Aggregation --> ConfidenceScore
ConfidenceScore --> CapabilityAssertion

15.5 Capability dependency graph

flowchart LR

Statistics --> MachineLearning
Python --> MachineLearning
DataCleaning --> MachineLearning
MachineLearning --> AIProductDevelopment
Communication --> ProductManagement
Prioritization --> ProductManagement
UserResearch --> ProductManagement
ProductManagement --> AIProductDevelopment

15.6 Capability transfer graph

flowchart LR

MilitaryLogistics --> SupplyChain
AcademicResearch --> MarketResearch
SoftwareEngineering --> QuantDevelopment
PoliticalScience --> GeopoliticalRisk
CivilEngineering --> InfrastructureConsulting
SupplyChain --> OperationsManagement

15.7 Capability decay model

flowchart TB

```
PriorConfidence --> TimeDecay
PriorConfidence --> PracticeDecay
PriorConfidence --> TechnologyObsolescence
PriorConfidence --> CertificationExpiry
PriorConfidence --> ContextChange
TimeDecay --> CurrentConfidence
PracticeDecay --> CurrentConfidence
TechnologyObsolescence --> CurrentConfidence
CertificationExpiry --> CurrentConfidence
ContextChange --> CurrentConfidence
RefreshedEvidence --> CurrentConfidence
```

15.8 Human review workflow

flowchart LR

```
ReviewRequired --> EvidencePacket
EvidencePacket --> Reviewer
Reviewer --> Decision{Decision}
Decision --> Approve[Approve]
Decision --> Downgrade[Downgrade]
Decision --> Reject[Reject]
Decision --> RequestMore[Request more evidence]
Approve --> ReviewEvent
Downgrade --> ReviewEvent
Reject --> ReviewEvent
RequestMore --> EvidenceRequest
ReviewEvent --> CapabilityUpdated
```

16. Constitutional Examples

16.1 Student completing an economics degree

| Step | Example |

|---|---|

| Canonical events | CourseCompleted, AssignmentGraded, ResearchPublished, CredentialVerified. |

| Evidence | econometrics exams, thesis, data analysis project, degree credential. |

| Ontology mappings | statistics, econometrics, policy analysis, research methods. |

| Graph mutations | Evidence nodes link to CapabilityObservations for statistical analysis and economic reasoning. |

| Reasoning | Strong assessed evidence supports academic economics capability; limited workplace evidence creates uncertainty. |

| Inference | CapabilityObserved: econometric_modeling at Strong confidence. |

| Confidence evolution | increases after thesis/project; decays if no recent analytical work. |

| Recommendations | market research internship, policy analyst role, data analytics bridge. |

| Passport updates | include verified degree and econometrics capability with evidence trail. |

16.2 Software engineer building open-source projects

| Step | Example |

|---|---|

Canonical events	GitHubRepositoryLinked, CommitObserved, PullRequestMerged, ProjectCompleted.
Evidence	merged PRs, tests, architecture docs, issue discussions.
Ontology mappings	software design, testing, collaboration, API design.
Graph mutations	repositories and artifacts evidence engineering capabilities.
Reasoning	Independent code evidence supports applied engineering; lack of employer review limits production confidence.
Inference	API design and testing capabilities Supported/Strong depending project quality.
Confidence evolution	strengthens with reviews, stars not treated as direct capability evidence.
Recommendations	backend role, open-source fellowship, production internship.
Passport updates	include selected repositories with contribution provenance.

16.3 Medical student entering residency

| Step | Example |

|---|---|

Canonical events	AssessmentGraded, ClinicalRotationCompleted, CredentialVerified.
Evidence	exams, OSCEs, supervised rotations, medical degree.
Ontology mappings	clinical reasoning, patient communication, procedural knowledge.
Graph mutations	supervised evidence links to clinical capability observations.
Reasoning	Regulated context requires verified credentials and jurisdiction rules.
Inference	clinical readiness capability only after supervised/verified evidence.
Confidence evolution	decays for procedures without recent practice.
Recommendations	residency pathways, remediation for weak rotations.
Passport updates	include regulated credential claims only if verified.

16.4 Researcher publishing papers

| Step | Example |

|---|---|

Canonical events	ResearchPublished, ResearchCapabilityMapped, ORCIDLinked.
Evidence	publications, datasets, citations, coauthor roles.
Ontology mappings	research design, methodology, domain knowledge.
Graph mutations	publication evidence supports research capability with authorship confidence.
Reasoning	Authorship contribution must be clear; citations are impact signals, not capability proof alone.
Inference	research design capability Strong if repeated and reviewed.
Confidence evolution	strengthens with independent publications and grants.
Recommendations	lab collaborations, grants, expert roles.
Passport updates	include selected publications and mapped research capabilities.

16.5 Construction worker gaining certifications

| Step | Example |

|---|---|

Canonical events	CredentialVerified, EmployerEvidenceVerified, ProjectCompleted.
Evidence	safety certificates, site supervisor review, completed projects.
Ontology mappings	safety compliance, equipment operation, site coordination.
Graph mutations	credential and employer evidence support regulated capability.
Reasoning	Certification expiry sharply affects current verified capability.
Inference	safety compliance capability Verified while credential current.
Confidence evolution	decays at expiry; refreshed by renewal.
Recommendations	renewal training, higher-responsibility roles.
Passport updates	include license/cert with expiry and verification status.

16.6 Military officer transitioning to civilian employment

Step	Example
Canonical events	ProfessionalExperienceObserved, CredentialVerified, EmployerInterested.
Evidence	service record, logistics projects, leadership reviews.
Ontology mappings	operations planning, logistics, leadership, risk management.
Graph mutations	military capabilities linked to civilian transfer hypotheses.
Reasoning	Transfer to supply chain is plausible but context gap remains.
Inference	logistics planning Strong; civilian supply-chain readiness Supported until direct evidence.
Confidence evolution	strengthens with civilian certifications/projects.
Recommendations	supply-chain certification, operations roles.
Passport updates	include transfer explanation and context limitations.

16.7 Entrepreneur building startups

Step	Example
Canonical events	ProjectCompleted, EmployerEvidenceVerified, PortfolioGenerated.
Evidence	product launches, revenue, customer interviews, team building.
Ontology mappings	product strategy, sales, leadership, financial planning.
Graph mutations	startup artifacts and outcomes support composite capabilities.
Reasoning	Outcomes matter but attribution and survivorship bias considered.
Inference	entrepreneurial execution capability Strong with validated outcomes.
Confidence evolution	strengthens across multiple ventures or reviewed outcomes.
Recommendations	venture roles, product leadership, incubators.
Passport updates	include projects and measurable outcomes.

16.8 Career switch from finance to AI

Step	Example
Canonical events	CourseCompleted, AssignmentGraded, ProjectCompleted, CareerIntentUpdated.
Evidence	statistics background, Python projects, ML coursework, portfolio.
Ontology mappings	finance analytics, statistics, Python, machine learning.
Graph mutations	finance capabilities connected to AI transfer hypotheses.
Reasoning	Statistical reasoning transfers; ML production evidence still required.
Inference	data analysis Strong; ML engineering Emerging/Supported depending projects.
Confidence evolution	strengthens with deployed AI projects and reviews.
Recommendations	ML portfolio projects, AI analyst roles, bridge curriculum.
Passport updates	include transferable analytics capabilities and AI learning evidence.

17. Governance

Governance area	Rule
Inference rule ownership	AI Platform Team owns implementation; Ontology, Graph, Reasoning, Security, and domain owners approve constitutional semantics.
Rule evolution	Rules are versioned and require regression tests before production.
Versioning	Inference models, confidence functions, evidence weights, prompts, and policy gates have independent versions.
Regression testing	Test sets cover evidence sufficiency, contradiction, decay, transfer, jurisdiction, bias, and revocation.
Bias monitoring	Evaluate outcomes by cohort, institution, country, language, accessibility, gender where legally/ethically permitted, and socioeconomic proxy where appropriate.
Fairness evaluation	High-impact inference must measure disparate error, false positives, false negatives, and

appeal outcomes. |

| Jurisdiction overlays | Local regulations affect verification/readiness, not global concept definitions. |

| Institution overlays | Institutional rubrics and mappings may specialize evidence, not redefine capability. |

| Policy overrides | Overrides publish canonical events and preserve prior history. |

Change process:

1. Propose inference rule/model change.
2. Identify affected capabilities, evidence types, graph paths, and products.
3. Run regression and fairness tests.
4. Review with Ontology, Graph, Reasoning, Security/Privacy, and domain owners.
5. Dual-run against prior version where high-impact.
6. Publish versioned release notes and migration/replay plan.
7. Monitor drift, appeals, and outcome quality.

18. Final Decision

INF-001 establishes how Syrka infers human capability. Capabilities are never declared by users, courses, LMSs, credentials, or AI models alone. They emerge through evidence-backed, ontology-grounded, graph-aware, confidence-scored, reviewable, reproducible inference.

This constitution governs capability inference, learning recommendations, career intelligence, employer matching, workforce planning, government analytics, and Job Passport generation. Every capability can be challenged, reviewed, strengthened, weakened, transferred, decayed, or revoked while preserving historical provenance and independence from any AI provider or LMS implementation.